

# Global Dental Workforce and Basic Oral-Health Service Availability

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Review proof — final Times New Roman export required.

## **The problem and why it matters**

WHO estimated that approximately 3.5 billion people were affected by oral diseases, with three out of four affected people living in low- and middle-income countries. The burden falls disproportionately on disadvantaged populations (WHO, 2022c). In 2022, the World Health Assembly adopted a global strategy envisioning universal health coverage for oral health by 2030 (WHO, 2022a, 2022c).

Dental care depends on trained labor. This paper asks how dentist density, measured as dentists per 10,000 people, is associated with basic oral-health service availability across countries at different income levels. Workforce supply, training barriers, financing, and incentives to serve underserved populations are the economic questions behind this comparison.

## **My interpretation of the evidence**

The analysis matches 171 countries using WHO dentist density, World Bank income groups, and three WHO 2021 service indicators: screening, urgent oral care, and basic restorative care (WHO, n.d.-b; World Bank, n.d.). The outcome measures general availability in public primary-care facilities, defined as reaching at least half of patients in need. Falling below that threshold does not mean no care exists (WHO, 2022b, p. 2).

The overall workforce–availability association was modest: Spearman rank correlation 0.306. Countries reporting all three services as generally available had a median density of 2.97, compared with 0.76 where none met that criterion and 0.72 where one or two did. The medians therefore did not rise steadily across categories (Figure 1; Appendix Table A1).

Income was closely associated with dentist supply. Median density rose from 0.06 in low-income countries to 0.31 in lower-middle-income, 2.33 in upper-middle-income, and 7.39 in high-income countries (Figure 2; Appendix Table A2). Within the two middle-income groups, density had almost no association with availability. This is consistent with income accounting for part of the overall pattern, but does not isolate its effect.

I repeated the analysis using each country's most recent dentist observation from 2021 or earlier, removing workforce observations from years after the service survey. The correlation barely changed, from 0.306 to 0.305, with all 171 countries retained. I am less concerned that the original inclusion of 2022–2023 observations explains the association.

Some workforce observations remain older than 2021. This check does not resolve reporting differences, current rather than historical income classifications, or the gap between availability and actual access. Workforce sources may count active or registered dentists and differ in completeness (WHO, n.d.-a). Cross-country associations cannot establish causation: income, health-system design, and spending could influence both variables. National density can also conceal local shortages. The analysis does not establish whether financing, deployment, or task sharing explains the differences (Appendix Table A3).

I expected dentist scarcity to be the dominant explanation. The findings made me question that expectation without establishing a better explanation.

Among the 99 countries reporting essential-curative benefit inclusion, those including it more often reported all three basic services as generally available (Table 1). The other 72 countries in the primary sample lacked this financing response.

**Table 1. Essential-curative benefit inclusion and basic service availability (2021)**

| Benefit included in<br>largest public scheme | Countries | All three services generally<br>available | 95% Wilson<br>interval |
|----------------------------------------------|-----------|-------------------------------------------|------------------------|
| Yes                                          | 78        | 49 (62.8%)                                | 51.7–72.7%             |
| No                                           | 21        | 9 (42.9%)                                 | 24.5–63.5%             |

Source: calculations from WHO 2021 benefit-package and public-primary-care service indicators (WHO, n.d.-b, n.d.-c). Intervals describe individual proportions, not their difference.

The two-sided Fisher exact test gives  $p = 0.135$ . This exploratory comparison is uncertain: patterns vary across income groups, several subgroups are small, and median

aligned dentist density is higher where the benefit is included (2.23 versus 0.72). Neither the test nor the intervals address income/workforce confounding or nonrandom reporting.

### **Policy options and my provisional judgment**

I am considering three options:

1. **Expand dentist training and retention.** This could increase capacity, but requires money and time; graduates may still choose affluent or urban areas.
2. **Improve deployment and primary-care integration.** This could help existing providers reach underserved populations, but requires ongoing recruitment, retention, referral, and public-sector support.
3. **Expand public financing for essential dental care.** This could reduce financial barriers, but increased demand may exceed provider capacity. The financing comparison makes it important to consider affordability and capacity together, without demonstrating a capacity constraint.

**Recommendation.** I recommend matching policy to workforce capacity and the barriers patients face, rather than adopting one approach everywhere. The income gradient, modest workforce association, and mixed financing results inform this provisional judgment; they do not establish a workforce threshold or rank policies. Where local evidence confirms severe provider shortages, I would prioritize training, retention, and deployment. Where providers can deliver care but affordability is a major barrier, I would prioritize financing with meaningful coverage, acceptable provider payments, and incentives to serve underserved areas.

**The strongest objection.** My cross-country evidence cannot establish that financing improves access. Benefit inclusion and availability may both reflect income or workforce capacity, and the data cannot show which came first. Romania includes essential-curative dental care in its largest public scheme, yet none of the three basic services is reported as generally available in public primary-care facilities (WHO, 2022b). Only 5% of dental

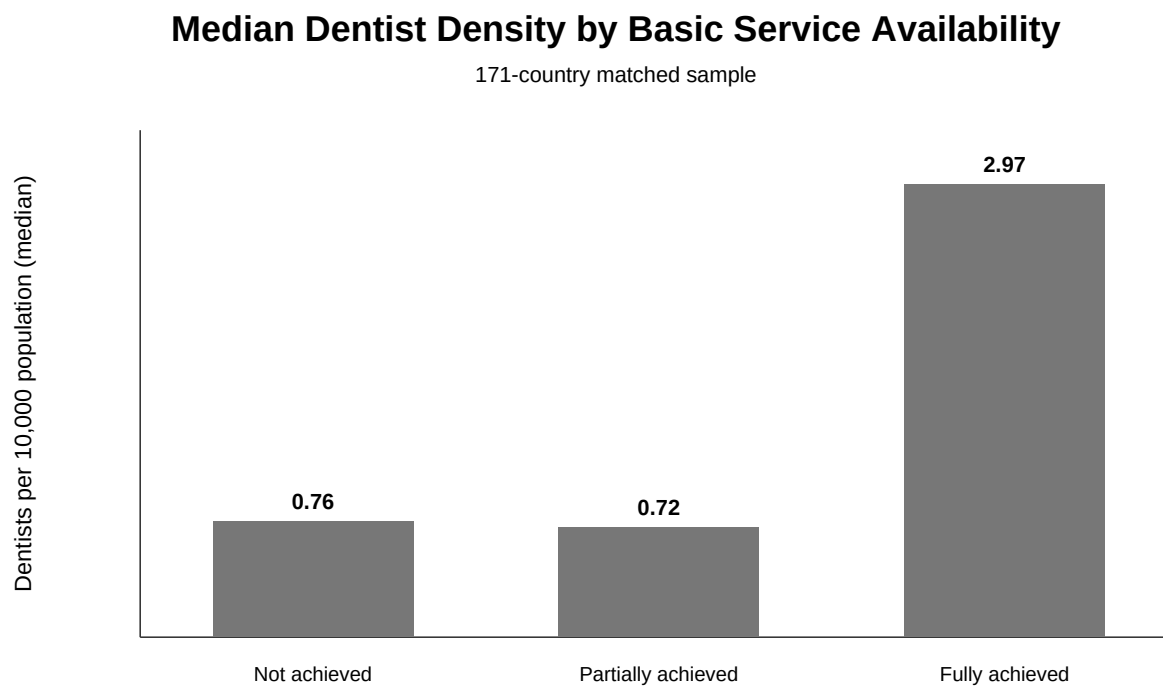
expenditure was publicly funded in 2021 (OECD/European Observatory on Health Systems and Policies, 2023, p. 15). This illustrates the distinction between entitlement and financial protection, without measuring a reform's effect. The spending share does not measure who receives care, and the public-facility outcome may miss private treatment. If supply cannot respond, subsidies could increase prices or waiting times; this is a theoretical risk, not an observed finding here.

**My response.** I accept the causal limitation. My recommendation is a conditional starting point, not a proven superior policy. Payments and cost-sharing rules should reduce patients' financial barriers while encouraging provider participation; low reimbursement can discourage participation, but has not been established as Romania's cause. Deployment incentives may be needed alongside financing. I would pilot reforms and evaluate treatment received, affordability, unmet need, and oral-health outcomes against costs. I would change my judgment if credible evaluations found no improvement after benefits expanded, or showed that workforce expansion or deployment performed better at comparable supply levels.

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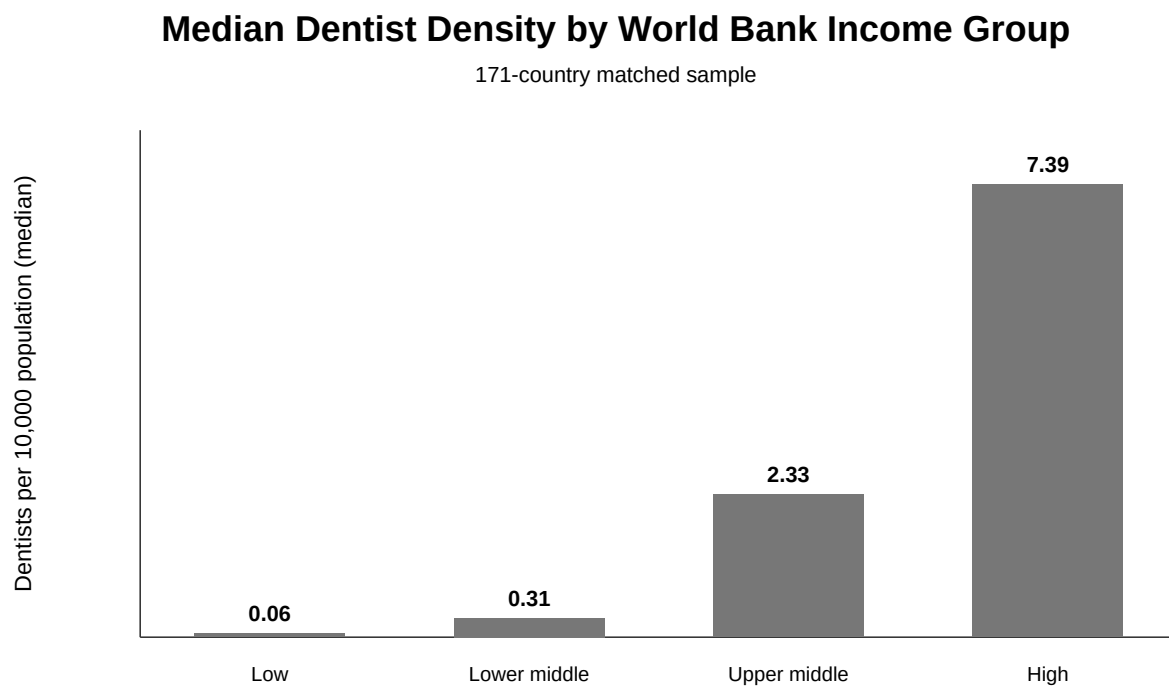
**Figure 1. Dentist density by service availability**



Source: WHO GHO/NCD CCS; World Bank. Author's calculations.

Categories count services generally available in public primary-care facilities. Medians use workforce observations through 2023 and WHO 2021 service components. Source: author's calculations from WHO GHO/NCD CCS and World Bank data. Median comparisons do not establish causation.

**Figure 2. Dentist density by income**



Source: WHO GHONCD CCS; World Bank. Author's calculations.

Income groups are current World Bank API classifications, rather than historical 2021 groups. Medians use workforce observations through 2023. Source: author's calculations from WHO workforce data and World Bank classifications.

## Appendix

**Appendix Table A1: Service availability and dentist density**

| Reported basic service availability            | Median dentists per 10,000 people |
|------------------------------------------------|-----------------------------------|
| None of the three services generally available | 0.76                              |
| One or two services generally available        | 0.72                              |
| All three services generally available         | 2.97                              |

Original workforce observations through 2023; outcome derived from WHO 2021 components. “Generally available” uses WHO's public-primary-care reach criterion.

**Appendix Table A2: Income group and dentist density**

| World Bank income group | Median dentists per 10,000 people |
|-------------------------|-----------------------------------|
| Low income              | 0.06                              |
| Lower middle income     | 0.31                              |
| Upper middle income     | 2.33                              |
| High income             | 7.39                              |

Source for Tables A1–A2: author's calculations from matched WHO workforce/service indicators and World Bank income classifications. These are original-baseline medians; the workforce-year sensitivity check is described in the main text.

### Appendix Table A3: Limits of the analysis

| What we cannot conclude                                         | Why                                                                                                                            |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| More dentists cause better service availability                 | The analysis shows associations, not cause and effect.                                                                         |
| Dentist supply does not matter                                  | A weak national relationship can conceal local shortages or differences in how dentists work.                                  |
| Financing, deployment, or task sharing explains the differences | These possible explanations have not yet been tested.                                                                          |
| Patients actually receive affordable, accessible care           | The outcome measures reported service availability in public primary care, not utilization, cost, travel distance, or quality. |
| A particular policy would improve access                        | The analysis has not evaluated the effects, costs, or tradeoffs of an intervention.                                            |

### AI assistance disclosure

ChatGPT assisted with explanations of the analysis and policy options, table formatting, and revisions to wording and structure. Some passages originated in AI explanations shared during the drafting process. The policy preference remains provisional, and further source verification and methodological checks are pending.